LESSON 6.3

# Explore Numerical Expressions with Exponents

**Set B · Version A** · Second form · supported practice

**MASTERY CHECK**
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can write and evaluate numbers in exponent form using a base and a power.

**WHY THIS MATTERS** You're a sound engineer at a music studio. Each time you flip a switch, the volume doubles. The starting volume is 1 unit. After flipping the switch 3 times, the volume is $2^3 = 2 \times 2 \times 2 = 8$ units — that's 8 times louder! How loud does it get after more flips?

REMEMBER

## Explore Numerical Expressions with Exponents

**Powers & Exponents. Formula: Base^Exponent (b^n = b · b · ... · b n times)** An exponent is a small raised number that tells you how many times to multiply a number by itself. We say $2^3$ as "two to the third power."

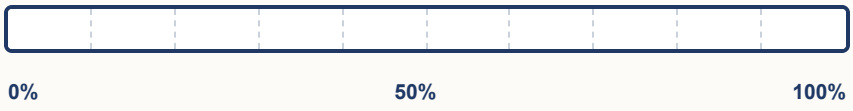
1. Base is the factor being multiplied repeatedly.
2. Exponent tells how many times to multiply the base by itself.
3. Evaluate: $2^3 = 2 \times 2 \times 2 = 8$ (not $2 \times 3 = 6$)

**Words:** Exponent *(Exponente)* · Base *(Base)* · Power *(Potencia)* · Evaluate *(Evaluar)* · Repeated multiplication *(Multiplicación repetida)*

**1** **Circle the letter of the best answer. Show how you know.**

12 is 25% of what number?

(A) 3 (B) 37
(C) 48 (D) 300




**WORK**

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

**2** **Circle the letter of the best answer. Show how you know.**

A team won 18 games, which is 60% of the games it played. How many games did it play?

(A) 10 games
(B) 28 games
(C) 30 games
(D) 48 games



0% 50% 100%

**WORK**

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

**3** **Circle the letter of the best answer. Show how you know.**

Find the whole: 15 = 0.30 × n. What is n?

(A) 0.02
(B) 4.5
(C) 45
(D) 50

**SHOW YOUR WORK**

**4** **Circle the letter of the best answer. Show how you know.**

9 students are 10% of a grade. How many students are in the grade?

(A) 0.9 students
(B) 19 students
(C) 90 students
(D) 900 students



0% 50% 100%

**WORK**

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

**5** Circle the letter of the best answer. Show how you know.

A music studio is recording tracks. Each track layer doubles the file size. A single track is 4 MB. The file size after n layers is given by $4 \times 2^n$. What is $2^5$?

(A) 10
(B) 16
(C) 25
(D) 32

**SHOW EACH PARTIAL PRODUCT, THEN ADD THEM**

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

**6** Circle the letter of the best answer. Show how you know.

A music studio is recording tracks. Each track layer doubles the file size. A single track is 4 MB. The file size after n layers is given by $4 \times 2^n$. How big is the file after 5 layers?

(A) 20 MB
(B) 40 MB
(C) 64 MB
(D) 128 MB

**SHOW EACH PARTIAL PRODUCT, THEN ADD THEM**

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

EXPLAIN YOUR THINKING

**How does the value change when you increase the exponent by 1? Why?**

**SENTENCE STARTER** When the exponent increases by 1, the value is multiplied by ___ because the base is used as a factor ___ more time.

**3-Point Exit Checklist:** □ 1. Units & Labels Verified □ 2. Mathematical Reasoning Stated □ 3. Calculations Double-Checked

**How did it go?** □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

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# Explore Numerical Expressions with Exponents

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**WHY THIS MATTERS** You're a sound engineer at a music studio. Each time you flip a switch, the volume doubles. The starting volume is 1 unit. After flipping the switch 3 times, the volume is $2^3 = 2 \times 2 \times 2 = 8$ units — that's 8 times louder! How loud does it get after more flips?

## PART 1 Understand the idea

models and meaning

### 1 Write the name of the correct group on each line.

Drag each repeated-multiplication beat next to its matching exponential form.

**Groups:** $7^3$ $6^2$ $4^4$ $9^2$

_____ $7 \times 7 \times 7$

_____ $6 \times 6$

_____ $4 \times 4 \times 4 \times 4$

_____ $9 \times 9$

### 2 Write the letter of the matching item on each line.

Match each power to the number of beats it plays (its value).

_____ $7^3$

_____ $6^2$

_____ $4^4$

_____ $9^2$

_____ $6^3$

_____ $7^2$

(A) 343

(B) 36

(C) 256

(D) 81

(E) 216

(F) 49

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

A music studio is recording tracks. Each track layer doubles the file size. A single track is 4 MB. The file size after n layers is given by $4 \times 2^n$. Why does the file grow so quickly?

(A) Because 4 MB is already large

(B) Because 5 is a big exponent

(C) Because the studio records faster each time

(D) Because each layer doubles the size instead of adding a fixed amount

SHOW YOUR WORK

**4** **Circle the letter of the best answer. Show how you know.**

The drum loop plays $3^4$ beats. What is the value of $3^4$?

(A) 7

(B) 12

(C) 27

(D) 81

SHOW YOUR WORK

**5** **Circle the letter of the best answer. Show how you know.**

A synth pattern repeats $7 \times 7 \times 7$. How do you write that as a power?

(A) $3^7$

(B) $7^3$

(C) $7 \times 3$

(D) $21^3$

SHOW YOUR WORK

**6** **Circle the letter of the best answer. Show how you know.**

The bass line doubles for $2^6$ counts. What is the value of $2^6$?

(A) 12
(B) 32
(C) 36
(D) 64

SHOW YOUR WORK

EXPLAIN YOUR THINKING

**How does the value change when you increase the exponent by 1? Why?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help